BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 1 - Report

Author: Tomáš Špelda

Date: 27. 3. 2026

Task 1

-

Task 2

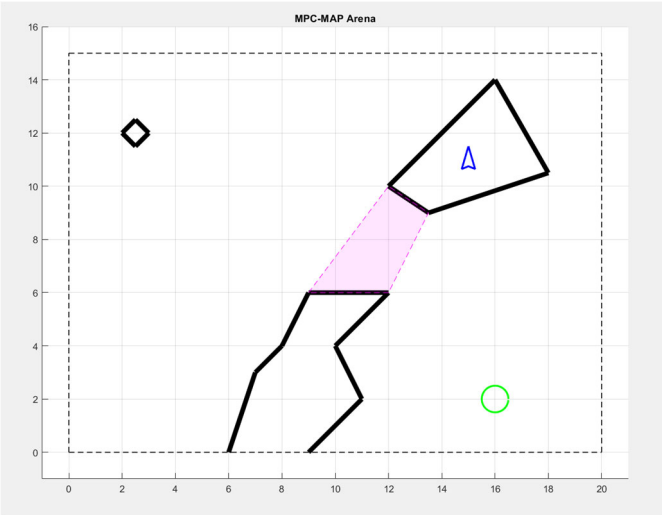


Figure 1 Used map

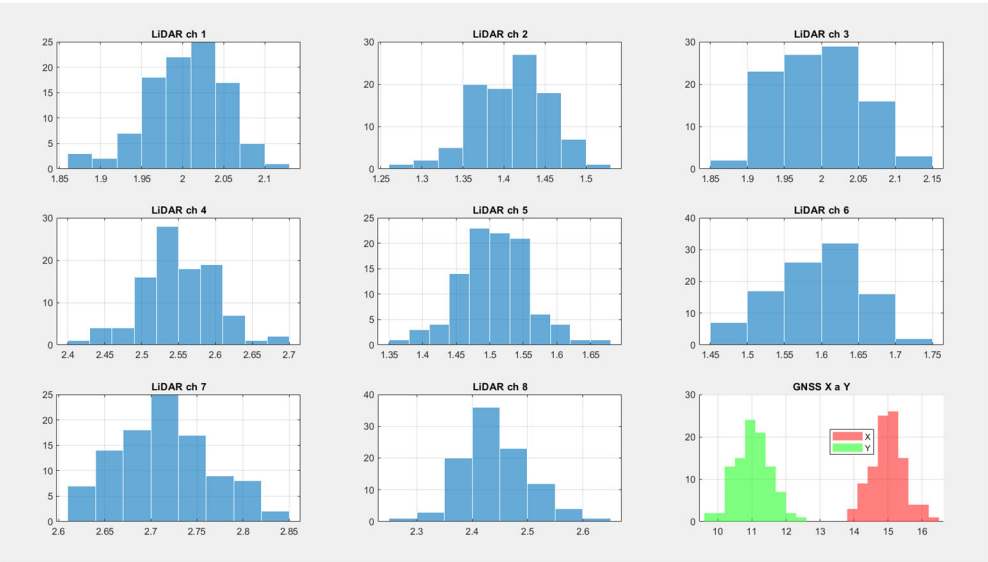


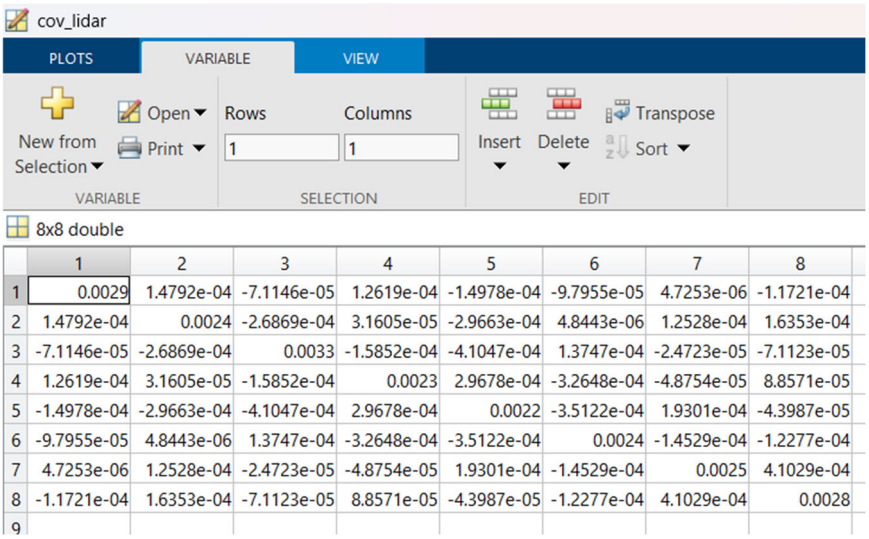
Figure 2 Histograms

The standard deviation is not consistent across individual LiDAR channels due to different distances from the walls.

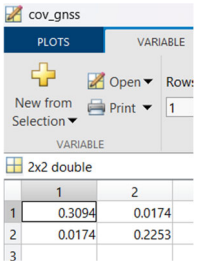
#

Task 3

The matrices have proper size and the values on the main diagonal are equal to sigma^2, see screenshot:

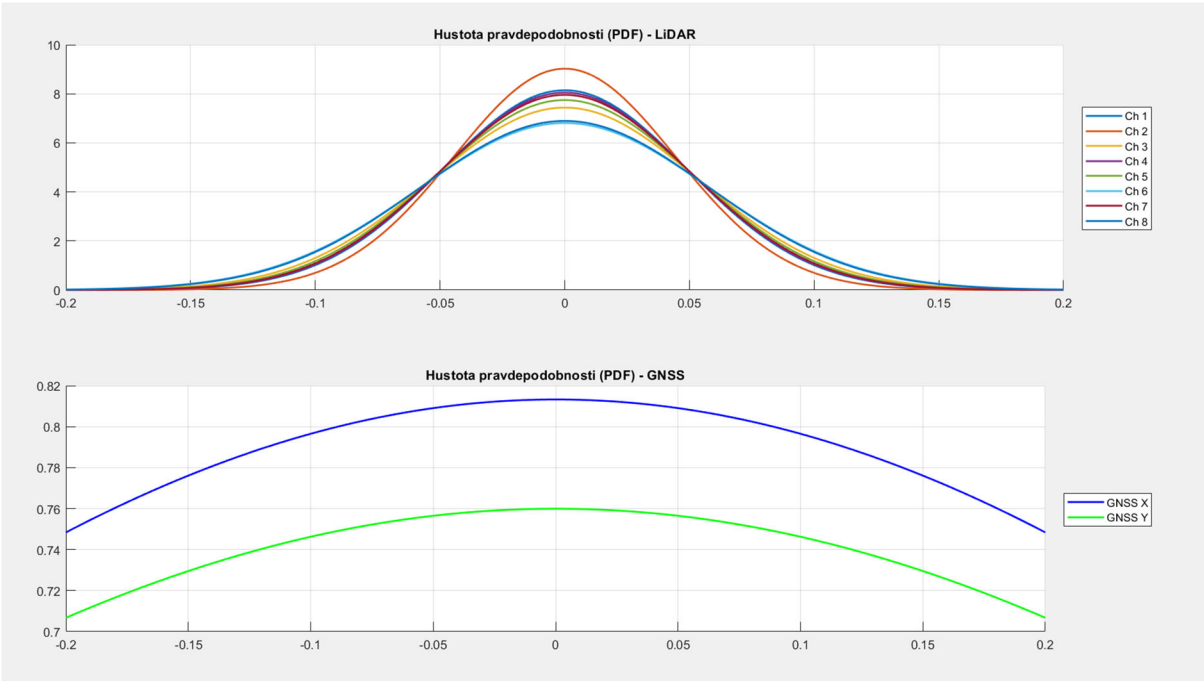


	1	2	3	4	5	6	7	8
1	0.0029	1.4792e-04	-7.1146e-05	1.2619e-04	-1.4978e-04	-9.7955e-05	4.7253e-06	-1.1721e-04
2	1.4792e-04	0.0024	-2.6869e-04	3.1605e-05	-2.9663e-04	4.8443e-06	1.2528e-04	1.6353e-04
3	-7.1146e-05	-2.6869e-04	0.0033	-1.5852e-04	-4.1047e-04	1.3747e-04	-2.4723e-05	-7.1123e-05
4	1.2619e-04	3.1605e-05	-1.5852e-04	0.0023	2.9678e-04	-3.2648e-04	-4.8754e-05	8.8571e-05
5	-1.4978e-04	-2.9663e-04	-4.1047e-04	2.9678e-04	0.0022	-3.5122e-04	1.9301e-04	-4.3987e-05
6	-9.7955e-05	4.8443e-06	1.3747e-04	-3.2648e-04	-3.5122e-04	0.0024	-1.4529e-04	-1.2277e-04
7	4.7253e-06	1.2528e-04	-2.4723e-05	-4.8754e-05	1.9301e-04	-1.4529e-04	0.0025	4.1029e-04
8	-1.1721e-04	1.6353e-04	-7.1123e-05	8.8571e-05	-4.3987e-05	-1.2277e-04	4.1029e-04	0.0028
9								

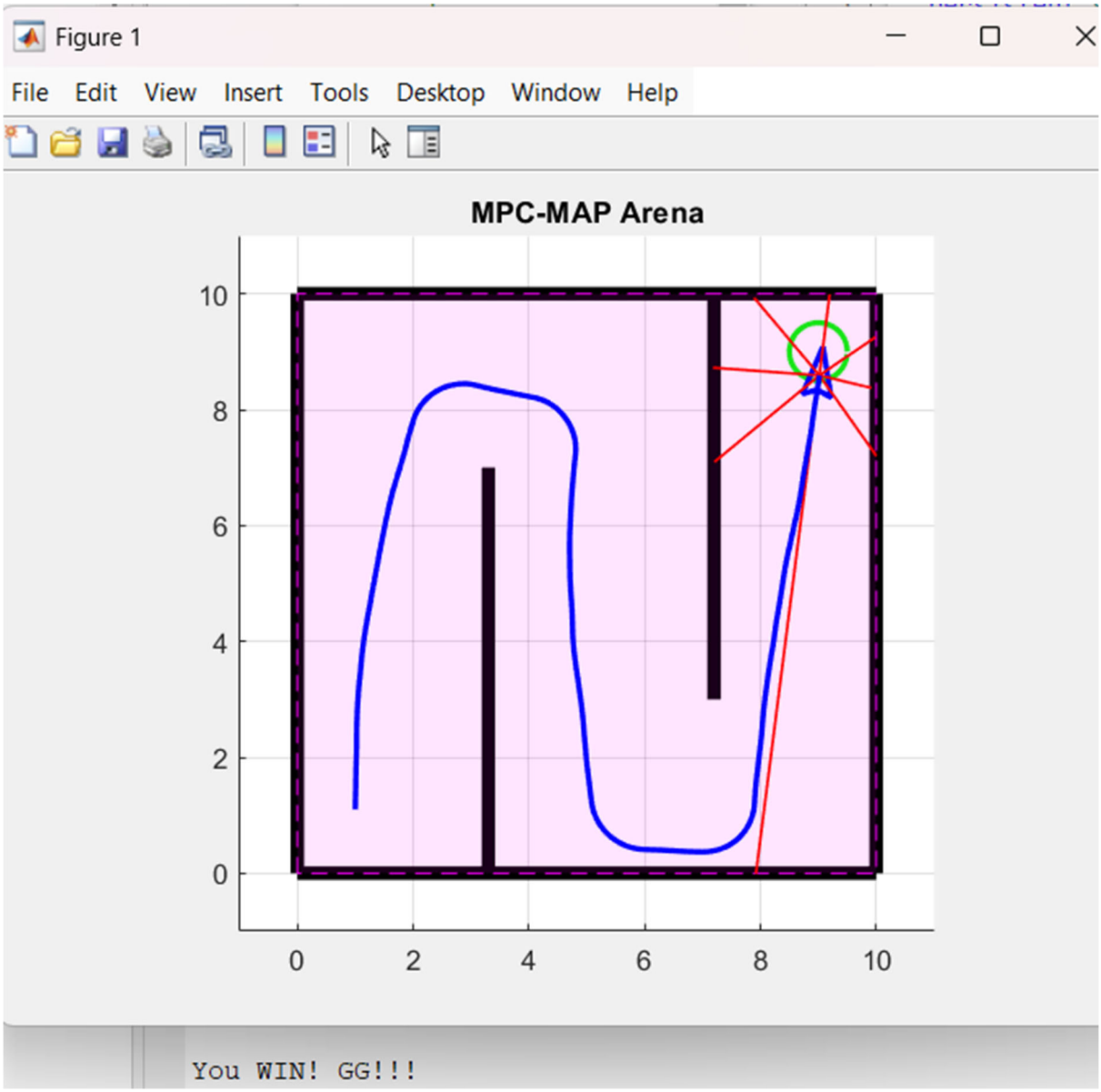


	1	2
1	0.3094	0.0174
2	0.0174	0.2253
3		

Task 4



Task 5



The uncertainty exist in the motion (grip, slip, radius of the wheel, distance of the wheels)

Github repository: https://github.com/Lagiontom/MPC-MAP